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Problem

- Classical RL relied on handcrafted state features and struggled to combine nonlinear function approximation with correlated online data.
- The challenge is one agent architecture that maps pixels to actions across visually different Atari games.

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Replay memory

- Each transition is stored in a replay buffer; random minibatches reuse past experience and mix many policies.
- Replay improves data efficiency and smooths the nonstationary stream seen by the neural network.

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State & network

PIXELS TO ACTION VALUES

- Frames are downsampled to 84x84 grayscale and four consecutive frames are stacked to expose motion.
- A convolutional network outputs one Q-value for every legal joystick action; epsilon-greedy behavior explores.

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Evidence

SEVEN ATARI 2600 GAMES

- One architecture and hyperparameter set is trained from pixels, reward and terminal signals with no game-specific features.
- It outperforms prior RL methods on six of seven games and exceeds a human expert on three.

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Learning rule

- $y = r + \gamma \max_a Q(s_{next}, a)$; $loss = (y - Q(s, a))^2$
- Backpropagation adjusts visual features and action values jointly from clipped rewards and Q-learning targets.

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Meaning & limits

- DQN established end-to-end deep reinforcement learning from raw sensory input and catalyzed modern deep RL.
- The early study covers few games, consumes many frames and can be unstable under bootstrapping and changing data.

DQN

Playing Atari with Deep RL

Mnih et al. | arXiv:1312.5602v1

Learn action values directly from raw game pixels by combining convolutional perception, Q-learning and replayed experience in one end-to-end agent.

Q-learning

experience replay

Atari pixels